

Post-Marketing Drug Safety Signal Detection Report

**A Disproportionality Analysis of CRESTOR (Rosuvastatin)
and Pneumonia Using FDA Adverse Event Reporting
System (FAERS) Data**

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Post-Marketing Drug Safety Signal Detection Report

Analysis type	Individual case safety report (ICSR) disproportionality analysis
Data source	FDA Adverse Event Reporting System (FAERS), 2026 Q1 public extract
Methodology	2x2 contingency table — Proportional Reporting Ratio (PRR) & Reporting Odds Ratio (ROR).
Prepared as	Independent portfolio project — Pharmacovigilance & Clinical Data Management skills demonstration
Report date	August 2026

This report was independently prepared to demonstrate applied skills in pharmacovigilance (PV) signal detection and clinical data management (CDM) using publicly available FAERS data, as part of professional certification and portfolio development. It is illustrative and not a validated regulatory submission.

Executive Summary

In this report I present an end-to-end pharmacovigilance signal detection exercise on a sample of U.S. FDA Adverse Event Reporting System (FAERS) 2026 Q1 quarterly data extract. Using standard disproportionality analysis tools (PRR and ROR), the objective was to assess for statistical disproportionality, a potential safety signal for the reporting pattern for the drug CRESTOR (rosuvastatin) and the adverse event Pneumonia.

The analysis revealed a Proportional Reporting Ratio (PRR) of 2.09, and a Reporting Odds Ratio (ROR) of 2.18, both of which were greater than the commonly used screening value of $PRR \geq 2.0$. The underlying case count was very small, however ($a = 1$ co-reported case), which is below the minimum case count criterion (commonly $N \geq 3$) that regulatory bodies like the MHRA require to prevent identifying signals due to chance in sparse data. The correct interpretation of this result is then that this is not, by itself, a validated safety signal, but rather a testament to the importance of using disproportionality measures alongside case-count and clinical plausibility measures, and not only for this reason.

Key figures at a glance:

2.09	13	36	1
PRR (signal ratio)	CRESTOR case-reports (unique cases)	Pneumonia case reports (unique cases)	Co-reported cases (drug + event)

Regulatory Recommendation: Continue routine surveillance; do not escalate to formal signal review based on this dataset alone. Re-run the analysis on the full, un-sampled FAERS quarterly extract to confirm case counts before any escalation decision.

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1. Research Question & Regulatory Context

1.1 Research Question

Is there a statistically disproportionate association between CRESTOR (rosuvastatin) and Pneumonia that is evidenced in real-world post-marketing spontaneous reporting data and warrants further pharmacovigilance evaluation?

1.2 Why Signal Detection

The inherent nature of RCTs is that they have a short time period and enroll a restricted, usually homogeneous, population. Rare adverse events, long latency effects, and effects related to real world comorbidities or polypharmacy effects are often only seen after a drug is prescribed to a much larger and more heterogeneous number of people after it has been approved. Pharmacovigilance (PV) is a process to constantly observe this real world experience of safety and one of the main data sources for PV are the databases of adverse events reported spontaneously.

1.3 Regulatory Framework

This analysis uses established pharmacovigilance principles for signal detection and disproportionality analysis:

U.S. FDA / FAERS: The U.S. Food and Drug Administration uses the FDA Adverse Event Reporting System (FAERS) for post-marketing safety surveillance. Disproportionality-based screening can be used to identify drug–event reporting patterns that warrant further evaluation.

ICH E2E: ICH E2E, Pharmacovigilance Planning, describes signal detection as an important component of ongoing pharmacovigilance activities and safety evaluation.

CIOMS Working Group VIII: Provides internationally recognized practical guidance on signal detection and the appropriate interpretation of disproportionality analysis alongside clinical assessment and other evidence.

Regulatory framing note

Disproportionality analysis is a **hypothesis-generating statistical screening tool**, not a confirmatory test of causality. A positive disproportionality result identifies a reporting pattern that may warrant further clinical and pharmacovigilance evaluation; it does not, by itself, establish that a drug causes an adverse event.

2. Data Source: FAERS Overview & Data Dictionary

2.1 About FAERS

The FDA Adverse Event Reporting System (FAERS) is a publicly available database that supports the FDA's post-marketing safety surveillance program. It contains adverse event and medication error reports submitted voluntarily by healthcare professionals and consumers, and mandatorily by pharmaceutical manufacturers. FAERS is released quarterly as a set of pipe-delimited flat files organized into related domains that must be joined on a common case identifier.

2.2 Domain Files Used in This Analysis

File/Domain	Description	Key Files Used
DEMO	Patient demographics and administrative case metadata (one row per case).	Primaryid, caseid, event_dt, age, sex, occr_country, rept_cod
DRUG	Drug/product information for every medication reported in a case (one row per drug per case).	primaryid, drugname, role_cod, dose_amt, dose_freq
REAC	Reported adverse event terms coded to MedDRA Preferred Terms (one row per event per case).	primaryid, pt (MedDRA Preferred Term)

Table 2.1 — FAERS domain files joined for this analysis; all joins performed on primaryid.

2.3 Data Dictionary Field Used In Merged Datasets

primaryid	DEMO / DRUG / REAC	Numeric	Unique identifier for an individual safety report; the join key across all domains.
caseid	DEMO	Numeric	Identifier for the underlying case; multiple primaryid versions can map to one caseid.
event_dt	DEMO	Date (YYYYMMDD)	Date the adverse event occurred, as reported.
init_fda_dt / fda_dt	DEMO	Date (YYYYMMDD)	Date the report was first / most recently received by FDA.
rept_cod	DEMO	Categorical	Report type, e.g. EXP (expedited), DIR (direct), PER (periodic).
drugname	DRUG	Text	Verbatim or standardized drug/product name as reported.
role_cod	DRUG	Categorical	Role of the drug in the case: PS (primary suspect), SS (secondary suspect), C (concomitant), I (interacting).
dose_amt / dose_unit / dose_freq	DRUG	Numeric Categorical	Reported dosing information (frequently missing/unstructured in real-world data).

Table 2.2 — Data Dictionary for Fields Used in the Merged Analytic Dataset

3. Data Quality Assessment

Before any signal detection statistics are calculated, a pharmacovigilance analyst must assess whether the underlying data is fit for purpose. The following checks were performed on the merged DEMO + DRUG + REAC dataset prior to analysis:

Check	Finding	Action / Implication
Record volume	10,000 rows sampled per source file (DEMO, DRUG, REAC) rather than the full quarterly extract, to keep processing lightweight for a portfolio-scale project.	Documented as a scope limitation (Section 9). Case counts in this report should be treated as illustrative of method, not as a production-scale signal read-out.
Join integrity	Inner joins on primaryid across DEMO → DRUG → REAC produced 470,471 drug × event rows. These records were then reduced to 963 unique primaryid-linked cases for the analytic sample. Because FAERS contains multiple drug and event records per report, row counts were distinguished from unique case counts when constructing the 2×2 contingency table..	Expected behavior — FAERS is a many-to-many structure by design; unique case counts (not row counts) were used for the 2x2 table.
Missingness	High missingness observed in dose_amt, dose_unit, dose_form, and lot_num / exp_dt fields, consistent with known FAERS data quality limitations (spontaneous, voluntary reporting).	These fields were excluded from the disproportionality calculation, which relies only on drugname, pt, and primaryid.
Drug/event name standardization	drugname values include brand, generic, and formulation-variant text (e.g., “SANDOSTATIN” vs “SANDOSTATIN LAR DEPOT”); pt values are pre-coded to MedDRA Preferred Terms.	A case-insensitive substring match (str.contains) was used for the target drug (CRESTOR) and event (PNEUMONIA); A production pipeline would additionally apply validated drug-name standardization, such as WHO Drug Dictionary (WHO-DD), to improve capture of brand, generic, formulation, and salt-form variants and reduce false matches or missed synonyms.
Duplicate case versions	FAERS allows multiple caseversion updates per caseid as follow-up information is submitted.	Not de-duplicated to caseid-latest-version in this sample; flagged as a limitation since it can inflate case counts in a full-scale run.

Table 3.1 — Data Quality Assessment of the Merged FAERS Analytic Dataset

4. CDM / SDTM Mapping

To demonstrate applied Clinical Data Management (CDM) knowledge, the FAERS legacy field structure was mapped conceptually to the corresponding CDISC SDTM domains that would be used to represent equivalent data within a clinical trial database. This mapping illustrates how a case management or data management professional would translate post-marketing safety data concepts into the standard clinical data model used across the drug development lifecycle.

FAERS Field(s)	SDTM Domain	SDTM Variable(s)	Mapping Rationale
primaryid / caseid (DEMO)	DM (Demographics)	USUBJID	Both serve as the unique subject/case identifier anchoring all other domain records.
age, sex, occr_country (DEMO)	DM (Demographics)	AGE, SEX, COUNTRY	Direct conceptual equivalents — baseline subject characteristics.
event_dt, pt (REAC)	AE (Adverse Events)	AESTDTC, AETERM / AEDECOD	pt is already MedDRA-coded, mapping directly to AEDECOD (dictionary-derived term); event_dt maps to the AE start date.
drugname, role_cod, dose_amt/unit/freq (DRUG)	EX (Exposure) / CM (Concomitant Meds)	EXTRT, EXDOSE, EXDOSU, EXDOSFRQ / CMTRT	role_cod = PS/SS routes the record to EX (suspect treatment) versus CM (concomitant medication, role_cod = C).
rept_cod (DEMO)	DM / Study metadata	(supplemental qualifier, no direct SDTM equivalent)	Report type has no clinical-trial analogue; retained as a PV-specific qualifier rather than force-mapped.

Table 4.1 — Conceptual Mapping from FAERS Fields to CDISC SDTM Domains

Standards Note In practice, post-marketing spontaneous report data and clinical trial data are governed by different data standards. FAERS is a post-marketing safety surveillance database, while ICH E2B(R3) provides the data standard for electronic Individual Case Safety Reports (ICSRs). Clinical trial data are structured according to CDISC standards such as SDTM.

5. Methodology

5.1 2x2 Table, PRR & ROR

Disproportionality analysis compares how often a specific drug and a specific adverse event are reported together, relative to how often each is reported with everything else in the database. This is organized into a standard 2x2 contingency table:

Drug / Event	Pneumonia (Event of Interest)	All Other Events
CRESTOR (Drug of Interest)	a — CRESTOR + Pneumonia	b — CRESTOR + Other Events
All Other Drug	c — Pneumonia + Other Drug	d — Neither CRESTOR nor Pneumonia

Table 5.1 — 2×2 Contingency Table for CRESTOR and Pneumonia

5.2 Formulas

Proportional Reporting Ratio (PRR):

$$\text{PRR} = [a / (a + b)] \div [c / (c + d)]$$

PRR compares the proportion of CRESTOR reports that mention Pneumonia to the proportion of all other-drug reports that mention Pneumonia. A PRR of 1 indicates no difference in reporting rate; regulatory screening conventionally treats $\text{PRR} \geq 2$ as a candidate signal worth further review.

$$\text{ROR} = (a \times d) / (b \times c)$$

ROR is the odds-ratio analogue of PRR, commonly used in EU pharmacovigilance practice (e.g., by the EMA), and tends to behave similarly to PRR for large, sparse safety databases.

SIGNAL DETECTION CRITERIA APPLIED

For this portfolio exercise, these criteria are used as a conservative screening framework: **PRR ≥ 2 , $\chi^2 \geq 4$, and N ≥ 3** . Meeting $\text{PRR} \geq 2$ alone is not sufficient to classify the finding as a formal signal. Results should be interpreted together with case volume, statistical precision.

6. Results

Case Counts

Metric	Value
Total unique cases in analytic sample	963
Cases with CRESTOR (drug of interest)	13 unique cases
Cases with Pneumonia (event of interest)	36 unique cases
a — CRESTOR + Pneumonia (co-reported)	1
b — CRESTOR + other events	12
c — Pneumonia + other drugs	35
d — Neither drug nor event	915

Table 6.1 — Case Counts and Disproportionality Results for the CRESTOR–Pneumonia Pair

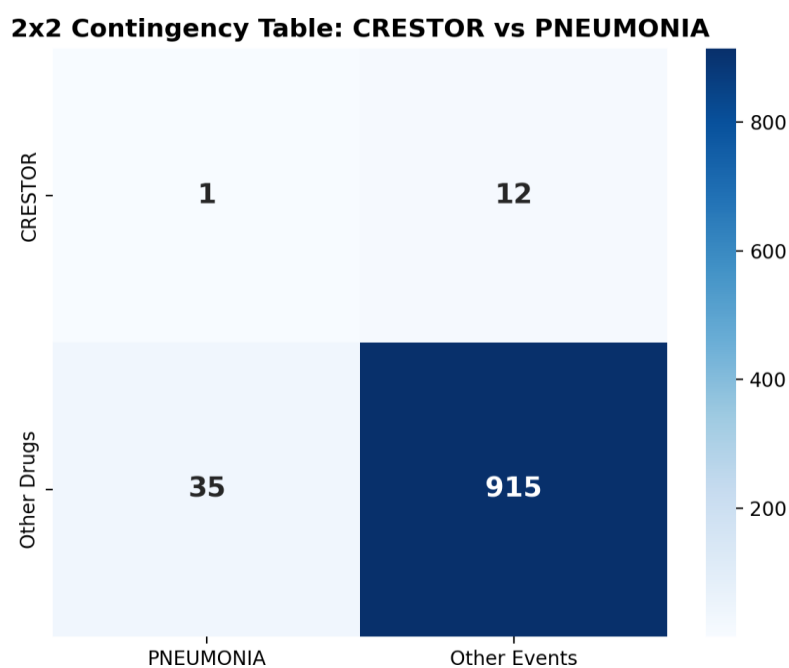


Figure 6.1 — Heatmap of Disproportionality Metrics for the CRESTOR–Pneumonia

Disproportionality Metrics

Metric	Result	Screening Threshold	Threshold Met?
PRR	2.09	≥ 2.0	Yes
ROR	2.18	Supportive; no fixed cutoff	Consistent with PRR
N (case count, a)	1	≥ 3	No

The CRESTOR–Pneumonia pair produced a **PRR of 2.09** and an **ROR of 2.18**, indicating a numerically elevated reporting proportion for the drug–event pair. However, only **one case (N = 1)** contributed to the observed combination, which does not meet the predefined minimum case-count criterion of $N \geq 3$. Therefore, the finding does not meet the complete screening criteria defined for this portfolio analysis and should be considered **hypothesis-generating rather than a validated pharmacovigilance signal**.

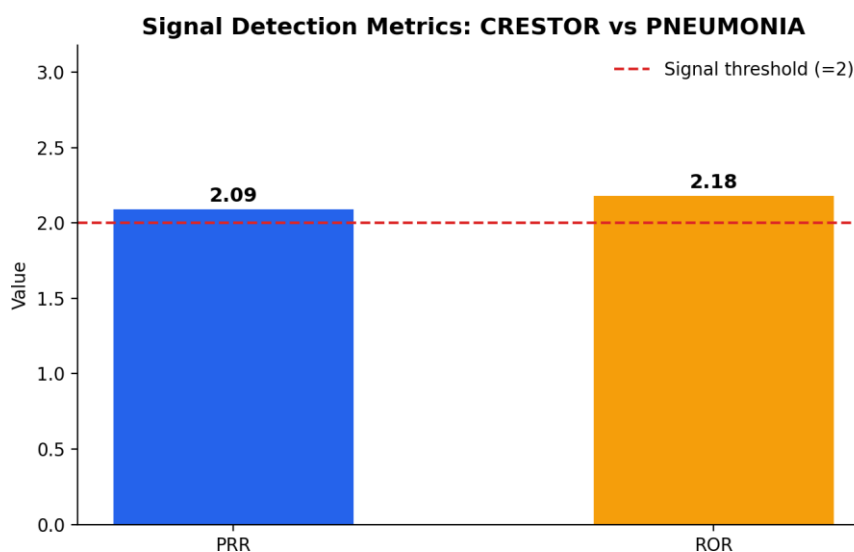


Fig:-6.2 PRR vs ROR values plotted against the conventional signal threshold of 2.0

Critical Statistical Caveat

PRR ≥ 2 is met, but **N ≥ 3** is not. Under the **multi-criteria screening framework defined for this portfolio analysis**, the result therefore does not meet the complete screening threshold because of insufficient case volume. A single co-reported case provides limited evidence for distinguishing a true disproportionate reporting pattern from a chance finding.

7. Clinical Interpretation

- From a purely numerical screening standpoint, the CRESTOR–Pneumonia pair shows a mild disproportionality signal, with a **PRR of 2.09** and **ROR of 2.18**. However, clinical interpretation must go beyond the numerical estimates and consider case volume, reporting characteristics, biological plausibility, and potential confounding.
- Case volume is too small to support a reliable conclusion. Only one co-reported case (N = 1) contributed to the CRESTOR–Pneumonia combination. With such a low case count, the disproportionality estimate is highly sensitive to random variation, and a single additional or missing report could substantially change the observed estimate.
- Biological plausibility should be assessed cautiously. The observed disproportionality does not establish a causal relationship between CRESTOR (rosuvastatin) and Pneumonia. Potential confounding, including underlying disease, comorbidities, concomitant medications, and differences in healthcare utilization, should be considered before drawing a causal conclusion.
- Reporting bias cannot be excluded. FAERS is a spontaneous reporting system and is subject to under-reporting, stimulated reporting, duplicate reporting, differences in reporting practices, and other reporting biases. Therefore, disproportionality measures should not be interpreted as incidence rates or measures of relative risk.
- The finding is hypothesis-generating rather than confirmatory. Although the PRR exceeds the commonly used screening value of 2, the observed N = 1 does not meet the predefined minimum case-count criterion of $N \geq 3$ used in this portfolio analysis. Therefore, the CRESTOR–Pneumonia pair does not meet the complete screening framework applied in this analysis.

Overall, the observed result warrants cautious interpretation but does not provide sufficient evidence to establish a confirmed pharmacovigilance signal or a causal association between CRESTOR and Pneumonia. Further assessment using a larger, appropriately deduplicated FAERS dataset and additional clinical and epidemiological evidence would be required before drawing stronger conclusions.

8. End-to-End CDM - Pharmacovigilance Data Flow

The diagram below illustrates how data flows from clinical trial data management, through post-marketing spontaneous reporting, into the signal detection and regulatory decision pipeline demonstrated in this report — showing how CDM and PV functions connect across the full product safety lifecycle.

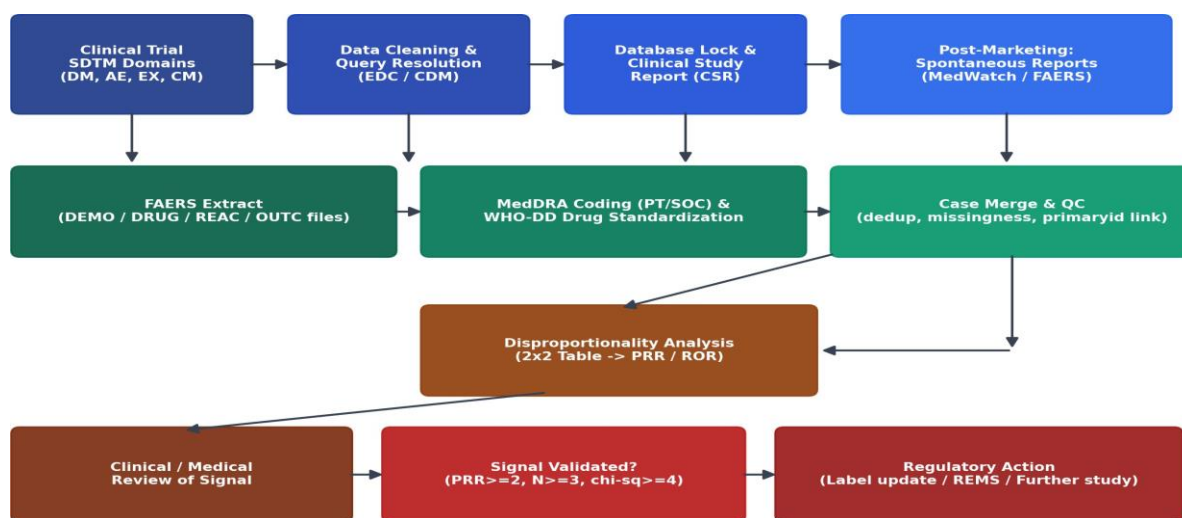


Figure 8.1 — Conceptual Data Flow from Clinical Trial CDM through Post-Marketing PV Signal Detection to Regulatory Action

9. **Limitations**

There are a number of limitations that have been documented with this exercise that should be taken into consideration prior to any production scale or regulatory use:

- Analysis of sampled, not full, data: A 10,000 row sample of each domain file of the FAERS data were analyzed instead of the entire quarterly extract. The reported number of cases is illustrative, and may not reflect the actual number of cases in the complete data set, therefore.
- No calculation of confidence intervals or chi-square analysis: This analysis only involved point estimates for PRR and ROR. A more extensive disproportionality analysis would also provide statistical measures like chi-square and confidence intervals to provide additional statistical precision and uncertainty.
- No case-version de-duplication: FAERS follow-up reports can have multiple versions of cases. Case counts for a full-scale analysis may be impacted by this analysis not narrowing the records to the latest case version.
- Partial validation of the drugs and events: Drug and event standardization was not performed fully using the validated list of drugs and MedDRA validation. This may result in missed synonyms or unintended matches.
- No risk adjustment for potential confounders: No stratification or adjustment was made for age, sex, reporting year, concomitant medications, or underlying conditions. These factors can affect spontaneous reporting behavior.
- One drug–event pair was analyzed: This portfolio exercise analyzed one drug–event pair (DEP) that was pre-defined. A production pharmacovigilance system would systematically use appropriate signal-detection methods to screen a much larger set of drug–event combinations.
- Small extent of drug-name standardization: In a production context, the use of validated drug-name terminology, e.g. WHO Drug Dictionary (WHO-DD), would be helpful in the improvement of identification of brand, generic and formulation/salt-form variants.

10. Conclusion & Regulatory Recommendation

Conclusion

This project illustrates a deliberately limited pharmacovigilance signal detection workflow, including FAERS data structure assessment, data quality evaluation, conceptual mapping of relevant data elements to CDISC SDTM, and calculation of standard disproportionality metrics (PRR/ROR). The CRESTOR–Pneumonia pair produced a PRR of 2.09 and an ROR of 2.18, exceeding the conventional PRR screening threshold of 2.0. However, only one co-reported case ($N = 1$) was observed, which does not meet the predefined minimum case-count criterion of $N \geq 3$ used in this portfolio analysis. Therefore, the finding should be considered **hypothesis-generating rather than an established pharmacovigilance signal** and should be further monitored and evaluated using a larger, appropriately deduplicated FAERS dataset.

Regulatory Recommendation

Recommendation	Rationale
Do not escalate to formal Signal Evaluation / PSUR discussion at this time.	$N = 1$ does not meet the predefined minimum case-count criterion ($N \geq 3$) used in this portfolio analysis.
Re-run the full analysis on the complete, un-sampled FAERS quarterly extract.	Current case counts are artifacts of a 10,000-row sample per file and are not representative of true reporting volume.
Add chi-square and 95% confidence interval reporting to the pipeline.	Point estimates alone are insufficient for regulatory-grade signal characterization.
Continue routine quarterly re-screening of this drug–event pair.	Standard PV practice for any borderline result, to detect any emerging trend over time.

Table 10.1 — Regulatory recommendation and rationale.

Disclaimer

This report was prepared as an independent, self-directed portfolio project to demonstrate practical skills in pharmacovigilance, FAERS data analysis, clinical data management, and signal detection methodology. The analysis uses publicly available FAERS data and is intended solely for educational and portfolio purposes. The dataset was analyzed using a scaled-down workflow and has not undergone the validation, quality controls, case-level medical review, or regulatory review required for production pharmacovigilance activities. Accordingly, the findings and recommendations should not be interpreted as evidence of a confirmed safety signal, causal association, or as a basis for clinical or regulatory decision-making.